Detecting Misinterpreted but Genuine Citations through Automated Claim Verification: A Review and Synthesis of Natural Language Processing Approaches to Citation Integrity

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Citations connect scientific claims to prior evidence, but bibliographic correctness does not guarantee citation fidelity. A paper may cite a genuine source while inaccurately paraphrasing its findings, omitting qualifications, broadening a result, reversing an effect, or drawing a conclusion that the cited evidence does not establish. Such errors are difficult to detect at scale because verification requires joint reasoning over a citing passage and the cited document, often across long, domain-specific texts. This review examines natural language processing (NLP) approaches for detecting such misinterpreted but genuine citations. We synthesize five related research threads: citation-function and citation-intent classification, citation sentiment and stance modeling, scientific claim verification, direct citation-integrity modeling, and retrieval-augmented large language models (LLMs) for scientific literature. We organize the literature using a taxonomy that separates citation existence failures from citation fidelity failures and distinguishes single-pair errors from network-level phenomena. Across the literature, a common technical pattern emerges: extract citation context, retrieve evidence from the designated source, verify the semantic relationship between claim and evidence, and optionally aggregate decisions over citation networks. Direct biomedical evidence is especially informative: Sarol et al. annotated 3,063 citation instances linked to 100 highly cited reference articles, with 39.18% containing accuracy errors; their best retrieval-plus-verification pipeline achieved 0.59 micro-F1 and 0.52 macro-F1. More recent work indicates that LLMs can contribute to reference-error detection and that retrieval-augmented systems can improve citation attribution, but these systems remain sensitive to retrieval quality, domain shift, context selection, and explanation reliability. The review concludes with research gaps in cross-disciplinary benchmarks, fine-grained labels, rationale evaluation, document-aware retrieval, robust LLM grounding, human-in-the-loop workflows, and network-level analysis.

CCS Concepts: • **Information systems** → **Information retrieval**; Digital libraries and archives; • **Computing methodologies** → *Natural language processing*; Artificial intelligence.

Additional Key Words and Phrases: citation integrity, citation accuracy, citation fidelity, scientific claim verification, evidence retrieval, natural language processing, large language models, scholarly communication

1 Introduction

Scientific writing depends on citations to establish provenance, situate contributions, justify methodological choices, and connect new claims to prior work. Citation counts and other bibliometric signals describe how often a work is referenced, but they do not by themselves establish whether a particular reference is used faithfully. A paper can cite a real article and still misrepresent it by omitting a key condition, broadening a limited result, converting correlation into causation, reversing the direction of an effect, or attributing a conclusion that the source does not actually support. These errors are particularly important because statements can be copied, paraphrased, and re-cited, allowing a local distortion to become embedded in a larger citation network.

Empirical studies have documented citation and quotation inaccuracies across scientific domains. Mogull's review and recalculation found that the estimated rate of quotation errors depends strongly on how studies are designed and classified, with major content errors representing a substantial fraction of identified inaccuracies [9]. Greenberg's analysis of a complete citation network illustrated how a speculative biomedical claim could accumulate apparent authority as later publications cited preceding articles without consistently returning to the underlying evidence [3].

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These studies establish the significance of the problem, but manual verification remains expensive because a reviewer must inspect both the citing statement and the referenced paper.

NLP offers a natural computational formulation. The system must identify the text expressing the citation-related claim, find relevant evidence inside the cited document, and assess whether that evidence supports, contradicts, partially supports, or fails to substantiate the claim. Several established NLP tasks address individual parts of this problem. Citation-function classification models infer why a work is cited [2, 11]; citation-sentiment methods estimate the rhetorical stance toward the cited work [7]; and scientific claim-verification systems retrieve evidence and classify support or refutation [12, 13]. A recent dedicated citation-integrity study combines these ideas directly for quotation-error detection in biomedical publications [10].

This review focuses on *misinterpreted but genuine citations*: cases in which the cited paper exists and is correctly identified, but the surrounding citing statement does not faithfully represent what the source establishes. The scope is narrower than general reference-error detection because metadata errors, missing references, and incorrect bibliographic records are treated as a separate problem. It is also narrower than open-domain fact checking because the intended source has already been selected by the citing author. Figure 1 makes this distinction explicit.

The paper pursues four questions. First, how can citation-integrity failures be decomposed into operational categories? Second, which NLP components contribute to automated fidelity checking? Third, what are the methodological differences among citation analysis, claim verification, direct citation-integrity systems, and LLM-based approaches? Fourth, which limitations in data, retrieval, verification, and evaluation currently constrain reliable deployment? To answer these questions, the review synthesizes representative empirical studies and uses their datasets, outputs, and modeling choices to derive a unified verification framework.

2 Review Scope and Method

2.1 Review objective

The objective is to connect adjacent research areas that are often studied separately. Traditional citation analysis describes the rhetorical or semantic role of references; scientific claim verification asks whether a scientific statement is supported by evidence; direct citation-integrity work asks whether a citation accurately represents a designated source; and retrieval-augmented LLMs address evidence-grounded synthesis over scientific literature. The central synthesis of this review is that citation-fidelity checking can be understood as a constrained form of evidence-grounded scientific claim verification in which the source document is known in advance.

2.2 Literature selection and organization

The review uses a focused narrative synthesis. Representative studies were selected because they provide either (i) foundational definitions for citation-function, citation-intent, sentiment, and citation analysis; (ii) public benchmarks for scientific evidence retrieval and verification; (iii) direct annotation and modeling of citation integrity; or (iv) recent retrieval-augmented LLM systems relevant to citation-grounded scientific synthesis. The synthesis emphasizes papers with identifiable datasets, explicit modeling procedures, reported evaluation criteria, and limitations that can be compared across task boundaries.

The organization follows four analytical dimensions:

(1) **Citation representation**: sentence-level citation context, citance, paragraph context, section structure, or a broader document representation.

(2) **Evidence access**: abstract-level evidence, sentence-level retrieval, full-document retrieval, or multi-document retrieval.
(3) **Decision mechanism**: rules and classical machine learning, neural or transformer classifiers, retrieval-plus-verification models, or LLM inference.
(4) **Output granularity**: citation function or sentiment, support/refute decisions, citation-integrity labels, error categories, or network-level risk signals.

These dimensions guide the literature synthesis in Section 3, the unified framework in Section 4, and the comparative analysis in Section 5. The verification decision itself is formalized in Equation (1), linking the extracted citation context and retrieved evidence to the final integrity judgment.

3 Literature Review and Related Work

3.1 Citation analysis and citation content

Citation analysis has traditionally emphasized counts, impact, and network structure, but full-text availability has enabled more fine-grained analysis of what authors actually say about cited work. Iqbal et al. reviewed a decade of NLP- and machine-learning-based in-text citation analysis and identified citation-context analysis, citation classification, citation sentiment, summarization, and citation-based recommendation as major computational directions [5]. This shift from metadata to citation content is a prerequisite for integrity checking because a fidelity judgment must inspect the linguistic relation between a claim and its source rather than merely observe that a citation exists.

A second literature reviews semantic citation classification itself. Kunnath et al. synthesized research on semantic classification of citations, including classification schemes, datasets, contextual features, classifiers, and evaluation protocols [8]. The broader lesson is that citation meaning is multidimensional: a reference can be used for background, methods, comparison, criticism, or other rhetorical purposes. Automated integrity checking therefore benefits from treating citation function as contextual information rather than assuming that every citation expresses the same kind of evidentiary claim.

3.2 Citation existence versus citation fidelity

Two dimensions are useful when discussing citation integrity. The first is *existence and bibliographic correctness*: whether the cited work exists and whether the reference record identifies it accurately. The second is *citation fidelity*: given a genuine source, whether the citing text accurately represents what that source says. These dimensions require different computational evidence. Metadata correction can often be handled with bibliographic databases or entity resolution, while fidelity requires semantic comparison between the citation context and the source document.

Within fidelity failures, some errors are local to a single citing–cited pair. Others emerge through propagation across a citation network. Greenberg's work demonstrates why network structure matters: when later papers repeatedly cite a prior claim without verifying its original support, the network can make the claim appear increasingly established [3]. A practical automated system should therefore preserve the distinction between pairwise citation accuracy and network-level amplification.

3.3 Citation function and citation intent

Before verifying a citation, a system can benefit from estimating what role the citation is playing. Teufel et al. introduced automatic classification of citation function, showing that citation contexts encode structured rhetorical information

beyond the presence of a reference marker [11]. Cohan et al. later introduced SciCite and showed how document-structure signals can improve citation-intent classification in scientific publications [2]. These tasks are relevant because the intended function changes the verification target. A citation used to report a borrowed method should be checked against methodological content, whereas a citation used to support a scientific proposition should be checked against findings and conclusions.

Citation-intent classification should, however, be treated as an auxiliary component. Knowing that a reference is used for comparison or background does not establish that the associated statement is accurate. The practical contribution of intent is better understood as task routing: it can determine which source sections, evidence types, and verification criteria deserve higher retrieval priority.

3.4 Citation sentiment and stance

Citation sentiment adds information about rhetorical polarity toward the cited work. Kilicoglu et al. compared rule-based and machine-learning approaches, including support-vector machines, convolutional neural networks, and bidirectional long short-term memory models, for citation sentiment classification in clinical research publications [7]. Such work demonstrates that citation-oriented language contains learnable stance signals.

Sentiment and fidelity are nevertheless different constructs. An author may accurately report a result and then criticize its limitations, while an inaccurate paraphrase may be expressed neutrally or positively. Consequently, sentiment should be used as contextual evidence rather than as an accuracy label. In a combined system, stance can provide a secondary signal about whether the source discussion may contain contrastive, critical, or qualifying language.

3.5 Scientific claim verification

Scientific claim verification addresses the semantic core of evidence checking. Wadden et al. introduced SciFact, a benchmark in which systems retrieve scientific abstracts containing evidence that supports or refutes a claim and identify rationale sentences for the decision [12]. The resource contains approximately 1.4K expert-written scientific claims paired with evidence-containing abstracts. Its relevance to citation fidelity lies in the joint retrieval-and-reasoning formulation: a claim should not be judged from lexical similarity alone; evidence must be found and interpreted.

MultiVerS extends scientific claim verification to weak supervision and full-document context [13]. The model was evaluated across several scientific claim-verification datasets, showing the usefulness of document-level context when evidence is not concentrated in a short abstract. For citation fidelity, the retrieval space becomes more constrained because the designated source is known, but the reasoning problem remains difficult because support can be distributed across methods, results, discussion, tables, and qualifying language.

The conceptual transition can be stated precisely. Open scientific claim verification asks, “Where in the literature is there evidence for this claim?” Citation-fidelity verification asks, “Does the source chosen by the author actually support the author’s statement about it?” The second problem reduces open-world retrieval but increases the importance of source-faithful interpretation.

3.6 Direct citation-integrity modeling

Sarol et al. provide a direct operationalization of citation integrity as an NLP task [10]. Their corpus contains 3,063 citation instances linked to 100 highly cited biomedical reference articles, and 39.18% of instances were labeled as containing accuracy errors. Annotators identified citation context in the citing paper, relevant evidence segments in the reference paper, and the citation-accuracy label. The authors modeled the problem with citation-context extraction,

BM25 retrieval with a MonoT5 reranker, and a fine-tuned MultiVerS verifier. Their best system reported 0.59 micro-F1 and 0.52 macro-F1 for citation-accuracy classification; their GPT-4 in-context experiments performed better on accurate citations than on erroneous ones.

The study is important because it shows both feasibility and difficulty. The problem can be represented as a complete pipeline, yet the errors of practical interest are often subtle. An evidence sentence can be highly similar to a citing statement while failing to preserve a condition, population, causal qualifier, or statistical interpretation. Retrieval therefore cannot be evaluated independently of verification, and verification cannot be interpreted independently of the evidence shown to the model.

3.7 LLMs and retrieval-augmented scientific literature systems

Recent LLM studies expand the set of tools available for citation verification. Zhang and Abernethy evaluated GPT-family models on an expert-annotated general-domain set of statement–reference pairs and varied how much source information was supplied through retrieval augmentation [14]. Their work reports that LLMs can identify erroneous citations with limited context and without fine-tuning, while also showing that model behavior depends on the reference information provided.

OpenScholar addresses a broader but closely related problem: retrieval-augmented scientific literature synthesis. Asai et al. report a scientific data store containing 45 million papers and an inference pipeline combining retrieval, reranking, generation, iterative self-feedback, and citation verification [1]. Although its primary task is literature synthesis rather than pairwise citation-fidelity classification, the architecture reinforces the value of explicit evidence retrieval and post-hoc citation verification. The shared design principle is that a model should ground scientific statements in identifiable source passages rather than relying solely on parametric memory. Two 2026 studies further connect LLM-based language understanding with direct citation-quality assessment. Isik and Guleryuzlu propose a two-stage LLM pipeline that extracts citation–reference pairs and assigns a semantic relevance score, with validation against expert judgments across engineering articles [6]. Huang et al. examine quotation-error detection with fine-tuned LLMs, alternative strategies for integrating full-text evidence, and TokenSHAP-based interpretability analysis [4]. Together, these studies broaden the methodological space from binary citation-error decisions toward continuous quality assessment and inspectable model behavior, while also highlighting the need to distinguish semantic relevance from complete source fidelity.

4 Taxonomy and Verification Framework

4.1 A working taxonomy of citation-integrity problems

Figure 1 organizes citation problems along two axes: whether the cited source is genuine and whether the problem can be localized to one citing–cited pair. Citation-existence failures require bibliographic resolution. Citation-fidelity failures require semantic comparison between the claim and its source. The single-pair branch contains the central target of this review, while network-level phenomena require aggregation across several citation relations.

The taxonomy also clarifies why citation counts alone cannot substitute for fidelity analysis. A highly cited paper may be repeatedly summarized inaccurately, while a low citation count may reflect disciplinary or discovery patterns rather than evidentiary quality. Pairwise semantic verification and bibliometric analysis therefore answer different questions and should not be conflated.

Table 1 gives an operational summary of the main fidelity-error categories used throughout the review.

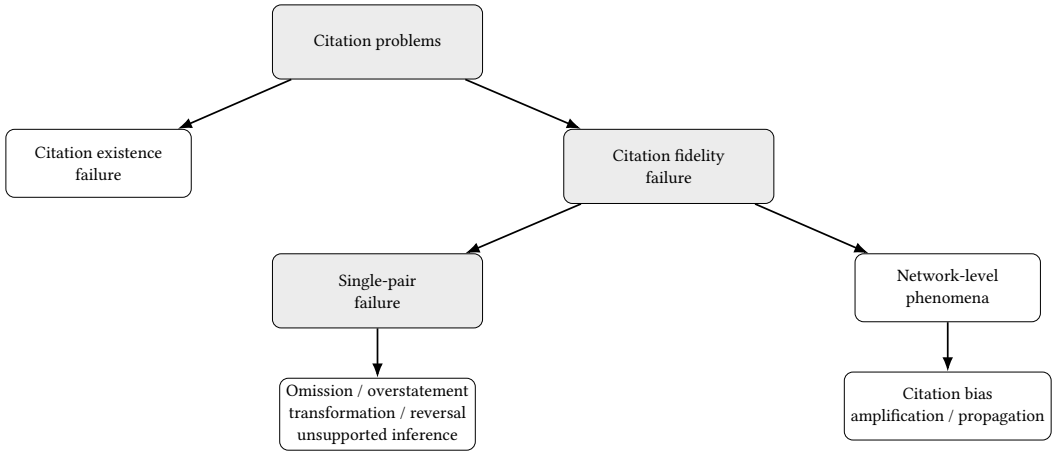


Fig. 1. Working taxonomy of citation problems. The review focuses on fidelity failures involving a genuine cited source, especially single-pair misinterpretation such as omission, overstatement, transformation, reversal, and unsupported inference. Network-level phenomena such as citation bias and amplification require additional aggregation over citation relations.

Table 1. Operational citation-fidelity error categories relevant to automated checking. Categories may overlap within one citation instance.

Error	Operational description	Verification evidence needed
Omission	A material qualifier, condition, population, limitation, or exception in the source is absent from the citing claim.	Source passages containing the omitted condition and the surrounding result or conclusion.
Overstatement	A limited, conditional, correlational, or preliminary result is presented as broader, stronger, more certain, or causal.	The original result plus methodological or statistical qualifiers that bound the source claim.
Transformation	The wording remains superficially related to the source while the semantic meaning changes.	Multiple source passages may be required to compare entities, scope, direction, or interpretation.
Reversal	The direction, polarity, or relationship described by the citing text conflicts with the source.	Source evidence that explicitly states the original direction or comparison.
Unsupported inference	The citing claim is plausible but is not established by the evidence in the cited work.	Evidence showing what is actually demonstrated, including relevant limitations and uncertainty.
Network amplification	A local mischaracterization is repeated or strengthened through later citations, creating apparent support at the network level.	Multiple citing documents, citation paths, and the underlying primary evidence.

4.2 A four-stage automated verification pipeline

A practical citation-integrity system can be decomposed into four stages, illustrated in Figure 2. First, *citation-context extraction* identifies the sentence containing the citation and any neighboring text required for interpretation. Second, *evidence retrieval* searches the designated reference article for passages relevant to the citation context. Third, *stance and accuracy verification* jointly interprets the claim and retrieved evidence. Fourth, an optional *network aggregation* stage combines pairwise decisions to study citation propagation, amplification, or selective support. Human review is positioned as an interpretive layer for flagged cases rather than as a replacement for the automated pipeline.

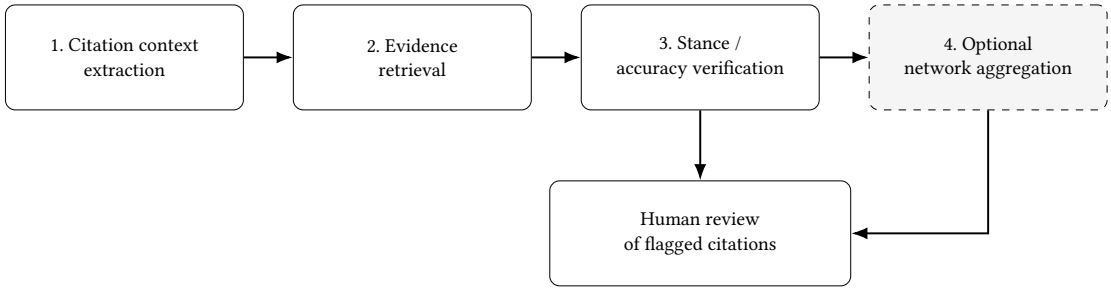


Fig. 2. Four-stage citation-integrity verification pipeline. The first three stages operate on the designated cited source; network aggregation is optional for propagation analysis, and human review can inspect flagged cases before editorial use.

Let c denote the citation context, $e_{1:k}$ the retrieved evidence passages, and r optional structural, rhetorical, or metadata features. A general verifier can be represented as

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} P(y \mid c, e_{1:k}, r), \tag{1}$$

where $\mathcal{Y}$ may contain labels such as {accurate, not accurate, irrelevant} or a finer-grained error taxonomy. Equation (1) emphasizes that citation verification depends on both evidence retrieval and contextual representation; a classifier cannot recover evidence that the retriever never surfaces.

4.3 Where each NLP component contributes

The components in Figure 2 are related but not interchangeable. Citation-context extraction determines the unit of analysis. Intent and sentiment can enrich the context representation but do not establish factual support. Retrieval determines which source passages are visible to the verifier. The verification model then estimates whether the claim is supported by the retrieved evidence, while network aggregation examines patterns that cannot be inferred from one pair alone.

This decomposition also suggests a diagnostic protocol. A false negative may be caused by poor context extraction, retrieval failure, or incorrect semantic verification. A false positive may arise when a retrieved passage is similar but not sufficient, when a caveat is ignored, or when an LLM generates an explanation that sounds plausible without being grounded in the designated source. Component-level evaluation is therefore necessary for interpreting end-to-end performance.

5 Comparative Analysis of Existing Approaches

The literature can be viewed as a progression from linguistic classification toward evidence-grounded, source-specific verification. Table 2 compares representative studies by task, methodology, dataset or scope, principal finding, and limitation for direct citation fidelity.

5.1 Shared design patterns

Three design patterns recur across the studies. First, citation context is a meaningful unit of analysis. A reference identifier without surrounding text does not reveal what proposition the author is attributing to the cited work. Second,

Table 2. Comparative synthesis of representative approaches relevant to automated citation-integrity verification. The studies address related but non-identical tasks, so the final column states the main transfer limitation when applied to citation fidelity.

Study	Task	Method	Dataset / scope	Main finding or contribution	Limitation for citation fidelity
Teufel et al. [11]	Citation function	Supervised classification of rhetorical citation roles using citation context.	Scientific articles; citation contexts.	Establishes that citation sentences encode structured rhetorical information.	Function classification does not establish whether the cited statement is faithful to the source.
Cohan et al. [2]	Citation intent	Structural scaffolds and multitask modeling; SciCite.	Multi-domain scientific publications.	Provides a larger citation-intent resource and incorporates document structure.	Intent is an auxiliary routing signal rather than evidence verification.
Kilicoglu et al. [7]	Citation sentiment	Rules, SVM, CNN, and BiLSTM comparison.	Clinical research publications.	Shows that stance and sentiment cues are learnable from citation contexts.	Positive, negative, or neutral stance is not equivalent to factual support.
Wadden et al. [12]	Scientific claim verification	Evidence retrieval plus support/refute classification with rationales.	SciFact; about 1.4K expert-written claims.	Establishes an evidence-grounded benchmark for scientific claims.	The source is not preselected; open retrieval differs from citation-specific checking.
Wadden et al. [13]	Full-document verification	Weak supervision and full-document context in MultiVerS.	Multiple scientific and health-related claim-verification datasets.	Demonstrates the value of long-document context for evidence reasoning.	Retrieval can still miss distributed, conditional, or domain-specific evidence.
Sarol et al. [10]	Citation integrity	Citation-context extraction, BM25 + MonoT5 retrieval, fine-tuned MultiVerS, and LLM baselines.	3,063 citation instances from 100 biomedical reference articles.	Provides a direct citation-integrity corpus and reports 0.59 micro-F1 / 0.52 macro-F1 for the best pipeline.	Biomedical scope, subtle errors, evidence-search difficulty, and moderate classification performance.
Zhang and Abernethy [14]	Reference-error detection	GPT-family LLMs under varying retrieval-context conditions.	Expert-annotated statement–reference pairs; general domain, with biomedical examples.	Shows that LLMs can detect some quotation errors with limited context and without fine-tuning.	Performance depends on context selection and model behavior; human oversight remains important.
Asai et al. [1]	Literature synthesis	Retrieval, reranking, LLM generation, iterative self-feedback, and citation verification.	45 million scientific papers in the OpenScholar data store; multidisciplinary benchmark.	Demonstrates the value of explicit retrieval and post-hoc citation verification for scientific synthesis.	The task is broader than pairwise citation-fidelity classification.
Isik and Guleryuzlu [6]	Citation quality scoring	Two-stage LLM pipeline for citation-sentence extraction and semantic relevance scoring, with expert validation.	121 engineering articles; 5,615 citation–reference pairs.	Extends automated citation assessment toward continuous quality scoring with expert-validated judgments.	Engineering-only scope and relevance scoring does not by itself encode all fine-grained source-fidelity errors.
Huang et al. [4]	Quotation-error detection	Fine-tuned LLMs with alternative full-text integration strategies and TokenSHAP interpretability analysis.	Quotation-error evaluation across source-text integration settings.	Studies both detection performance and interpretability of LLM-based quotation-error models.	Results remain sensitive to how source text is represented and integrated, motivating broader cross-domain evaluation.

retrieval is necessary whenever the evidence is not contained in a short abstract or obvious section. Third, the claim and evidence need to be interpreted jointly, whether by a feature-based classifier, transformer model, or LLM.

A fourth pattern becomes clear when direct citation-integrity systems are compared with general claim verification: the verification target is not simply semantic similarity. Similarity can identify a topical match while missing qualifiers or methodological restrictions. Conversely, lexical divergence does not imply contradiction. This motivates explicit evidence rationales and source-grounded explanations rather than a single opaque score.

5.2 Methodological trade-offs

Rule-based approaches are comparatively transparent and can encode domain conventions, but they become brittle when wording varies. Classical machine-learning models are efficient for narrow tasks but depend on manually selected features and annotation quality. Transformer-based methods improve semantic matching and contextual representation, but full-document input increases inference and retrieval complexity. LLMs provide flexible reasoning and can operate with little task-specific training, yet their behavior is sensitive to prompt design, retrieved context, context length, and model version.

The literature therefore supports a hybrid architecture in which retrieval constrains the evidence space, a verifier makes the factual decision, and an LLM is used selectively for interpretation or explanation. This is a design synthesis rather than a claim that any single family is sufficient. In particular, the direct citation-integrity results in Sarol et al. show that evidence retrieval and subtle error recognition remain difficult even when strong neural components are combined [10].

6 Evaluation Considerations

6.1 Annotation quality and evidence localization

Citation-integrity annotation is difficult because annotators must understand both the citation context and the source article and must identify which parts of the source constitute sufficient evidence. Sarol et al. report challenges in locating evidence in long full-text articles and in achieving consistent evidence-segment annotation [10]. Their annotation design allowed relevant evidence to be represented as sentences, sections, or paragraphs, reflecting the fact that support is not always localized to one sentence.

A robust benchmark should therefore report not only the final citation-accuracy label but also the annotation protocol, evidence spans, agreement measures, and error category definitions. Without these details, it is difficult to determine whether a model is learning genuine citation reasoning or exploiting shallow textual cues.

6.2 Retrieval evaluation and end-to-end verification

Retrieval and verification should be evaluated separately and jointly. Retrieval metrics such as recall@k and mean reciprocal rank indicate whether relevant evidence is entering the verifier's context. End-to-end measures such as precision, recall, and macro-F1 then describe whether the final citation-integrity labels are correct. Macro-F1 is useful when inaccurate citations are less frequent than accurate citations because it gives more visibility to minority classes.

Rationale quality should be evaluated as a third dimension. For a scholarly workflow, a label without inspectable evidence is difficult to audit. A useful evaluation protocol should ask whether the cited passage is actually sufficient for the decision, whether the rationale refers only to retrieved evidence, and whether important qualifiers are preserved.

6.3 Generalization across scientific domains

Direct citation-integrity evidence is still concentrated in biomedical literature, most notably the corpus of Sarol et al. [10]. This matters because scientific writing conventions differ across fields. Biomedical claims often rely on explicit populations, interventions, outcomes, and statistical qualifiers; physics may rely on equations, theoretical assumptions, or simulation conditions; computer science papers may describe benchmarks, model configurations, and datasets; and the humanities may require interpretation of extended textual context.

A model that performs well within one domain should therefore not automatically be treated as a general-purpose verifier. Cross-domain evaluation should test both in-domain performance and transfer to unseen disciplines, article types, and claim styles.

6.4 Robustness to long context and incomplete access

Source accessibility is another practical constraint. Some reference articles are openly available in machine-readable form, while others may be available only through publisher interfaces or imperfectly extracted PDFs. Evidence retrieval can fail because relevant information is omitted from an accessible version, split across sections, embedded in a figure or table, or expressed through an equation. Long-context LLMs do not remove this problem: OpenScholar's analysis explicitly shows that the number and quality of retrieved passages affect both correctness and citation accuracy [1].

This suggests that robust systems need document parsing, structural retrieval, and explicit handling of incomplete evidence. A verifier should be able to distinguish "not supported by the retrieved evidence" from "not supported by the source," because the first may simply reflect a retrieval failure.

7 Research Gaps and Future Directions

7.1 Larger and more diverse annotated benchmarks

The most immediate bottleneck is high-quality labeled data. Direct citation-integrity annotation is expensive because experts must read across documents and resolve subtle semantic distinctions. Future benchmarks should include multiple disciplines, article genres, citation functions, error severities, and source-document structures. They should also preserve the exact evidence spans used by annotators so that retrieval and verification can be evaluated independently.

The existing trajectory suggests a complementary benchmark design. SciFact provides expert-written claims and evidence rationales [12], Sarol et al. provide source-specific citation-integrity annotations [10], and Zhang and Abernethy provide general-domain statement–reference pairs for LLM-based reference-error detection [14]. Recent work also contributes expert-validated citation-quality scoring and interpretable quotation-error analysis [4, 6]. A unified benchmark could preserve the strengths of these resources while adding cross-disciplinary evaluation.

7.2 Fine-grained labels and rationale-centered outputs

Binary labels are insufficient for many editorial applications. A useful system should distinguish omission, overstatement, transformation, reversal, unsupported inference, and other categories when evidence allows. These labels are not merely descriptive: they determine what a human reviewer needs to inspect.

Rationale production should therefore be treated as part of the core task. An effective alert can provide the original citation context, one or more source passages, the predicted relation, and a concise explanation of the mismatch. Such outputs transform verification from an opaque classification problem into an auditable evidence-comparison process.

7.3 Document-aware retrieval beyond lexical similarity

BM25 and neural rerankers offer practical retrieval baselines, but topical similarity is not equivalent to evidentiary sufficiency. Future retrieval systems should incorporate article structure, section semantics, study design, qualifiers, tables, figures, equations, and cross-sentence dependencies. For example, a claim about statistical significance may require retrieval of both a result sentence and the associated analysis or methodological qualification.

Hybrid retrieval is also promising. A lexical retriever can recover terminology-sensitive passages, while a dense retriever can identify semantically related evidence and a cross-encoder can rerank the candidates. The key research question is not merely whether a retriever finds a similar passage, but whether it finds enough of the source context to support a defensible verification decision.

7.4 LLM verification with explicit grounding

LLMs can contribute to nuanced evidence interpretation, but the literature supports grounding them in retrieved source material and evaluating how explanations are produced. Zhang and Abernethy show that model performance changes as reference information is varied [14]; OpenScholar similarly reports improvements from retrieval, reranking, iterative feedback, and citation verification [1]. Isik and Guleryuzlu add expert validation to an LLM-based relevance score, while Huang et al. examine interpretability for quotation-error detection [4, 6]. These results motivate architectures that explicitly separate retrieval, evidence selection, decision, explanation, and uncertainty.

A grounded verifier should also expose uncertainty. When evidence is incomplete or conflicting, the system should not be forced into a high-confidence binary judgment. Calibrated uncertainty, abstention, and escalation to human review are especially important in high-impact scholarly workflows.

7.5 Human-in-the-loop editorial workflows

Automated verification is best positioned as a screening and triage layer. Authors, reviewers, and editors can inspect flagged citations, verify the highlighted source passages, and decide whether the mismatch is substantive. This design avoids treating model predictions as final scientific judgments and provides a practical path for deployment even when model performance is imperfect.

Human-in-the-loop evaluation should measure more than model accuracy. It should examine whether alerts save reviewer time, whether highlighted evidence is sufficient, whether false positives create avoidable burden, and whether users can understand the reason for a flag. These human-centered measures are likely to determine the practical value of citation-integrity systems.

7.6 Network-level detection of propagated distortion

Single-pair verification cannot capture all forms of citation distortion. Greenberg's network analysis illustrates how unsupported ideas may become more authoritative when repeated through a citation chain [3]. Future systems could combine pairwise fidelity decisions with citation-network structure to identify claims that are repeatedly cited despite weak primary evidence.

Network aggregation should remain a distinct output from pairwise accuracy. A citation may be locally faithful even when a broader network exhibits selective visibility or amplification, and a single inaccurate citation may not imply a network-level distortion. Keeping these levels separate supports more precise diagnosis and evaluation.

7.7 Cross-disciplinary and multimodal evidence

Many scientific claims depend on more than prose. Tables, plots, equations, supplementary material, and structured metadata can contain the evidence needed to interpret a citation correctly. Future citation-verification systems should therefore move toward multimodal document understanding, particularly for fields in which numerical results or mathematical derivations are central to the cited claim.

This direction also raises new evaluation requirements. A model should be tested on cases where the decisive evidence occurs outside ordinary paragraph text and on cases where extracted PDF structure is imperfect. Such benchmarks would reveal whether a system truly verifies scientific evidence or merely compares nearby textual passages.

8 Discussion

The synthesis in this review suggests that citation-integrity verification is not a single classification problem. It is an evidence-grounded workflow whose quality depends on the interaction of context extraction, retrieval, semantic verification, and explanation. This perspective reconciles findings across adjacent literatures.

Citation-intent research establishes that the citing sentence contains meaningful rhetorical information [2, 11]; citation-sentiment work shows that stance can be learned from local cues [7]; SciFact and MultiVerS demonstrate the importance of retrieval and evidence reasoning for scientific claims [12, 13]; and direct citation-integrity research shows that these components can be assembled into a source-specific verification pipeline [10]. Recent LLM research adds flexible reasoning, but also increases the need for explicit grounding, citation-quality assessment, and verification [1, 4, 6, 14].

The resulting design principle is therefore modularity with auditable interfaces. Each stage should expose intermediate artifacts: the extracted citation context, retrieved passages, final evidence set, predicted relation, and rationale. This makes errors diagnosable and allows future improvements to be evaluated at the component level. It also reduces the risk that a fluent model-generated explanation is mistaken for evidence.

The literature further suggests that "citation accuracy" should not be treated as a single universal label. The useful output depends on the decision context. For an author preparing a manuscript, a simple warning with source evidence may be sufficient. For an editor screening a high-volume submission stream, a ranked triage list with calibrated uncertainty may be more useful. For bibliometric research, network-level aggregation may matter more than individual alerts. A flexible system should therefore separate the core evidence-verification layer from downstream workflow-specific outputs.

9 Conclusion

Misinterpreted but genuine citations occupy an important gap between bibliographic correctness and full scientific truth assessment. The reference may be real and correctly identified while the citing statement nevertheless distorts the source through omission, overstatement, transformation, reversal, or unsupported inference. The reviewed literature shows a clear methodological connection between citation analysis, scientific claim verification, direct citation-integrity modeling, and retrieval-augmented LLM systems.

The central framework developed in this review is a four-stage pipeline: citation-context extraction, evidence retrieval, semantic accuracy verification, and optional network-level aggregation, with human review for consequential decisions. Equation (1) formalizes the verifier as a decision over the citation context, retrieved evidence, and auxiliary features, while Figures 1 and 2 and Tables 1 and 2 provide complementary views of the problem and the existing literature.

Direct evidence from Sarol et al. demonstrates that automated citation-integrity checking is feasible but still difficult: their 3,063-instance biomedical corpus contains substantial citation errors, and their best retrieval-plus-verification pipeline achieves moderate F1 performance [10]. LLM-based studies add useful capabilities, but they likewise reinforce the importance of retrieval, evidence selection, source-grounded verification, and interpretable quality assessment

[1, 4, 6, 14]. Future progress therefore depends on larger cross-disciplinary datasets, fine-grained error labels, rationale-centered evaluation, document-aware retrieval, multimodal evidence handling, calibrated uncertainty, and human-in-the-loop editorial workflows. These directions position automated citation verification as a practical layer of infrastructure for more transparent and trustworthy scholarly communication.

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